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Designing hope-oriented service touchpoints for adolescent oral health in public dental care

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ABSTRACT: Adolescence is a critical period for establishing oral health habits, yet motivating young people remains challenging. This paper presents a pilot intervention in the Norwegian Public Dental Service, developed through co-design and guided by salutogenic and hope theory. The intervention consisted of three touchpoints: an SMS invitation, an in-clinic goal-setting dialogue, and a follow-up SMS embedded in routine care. Based on interviews with clinicians and adolescents, we explore how hope-oriented communication can be enacted within everyday clinical practice.

KEYWORDS: service-oriented design, healthcare design, co-design, human-centred design, communication

1. Introduction

Adolescence is a critical period for establishing lifelong health habits, including oral health (Nazari et al., 2025; Tsai et al., 2020). Although population-level indicators have improved in Norway and other high-income countries with publicly organized dental care for children and special needs groups, a substantial proportion of young people still experience dental caries, and social inequalities persist (Høvik et al., 2025; Statistics Norway, 2023). While service design is increasingly recognized as a driver of innovation in healthcare, its application within oral health, particularly in adolescent health promotion, is still limited (Leason et al., 2024).

Understanding how to motivate adolescents to care for their oral health in everyday life remains a central concern for public dental services (Høiseth et al., 2024; Tsai et al., 2020). Oral health promotion involves coordinated efforts across policies, environments, community initiatives, personal skill development, and service adaptation, encompassing interventions at individual, community, and professional levels (Peerbhay et al., 2025).

Systematic reviews suggest that theory-based oral health interventions can improve knowledge, self-efficacy, behavior, and short-term clinical outcomes among children and adolescents (Nazari et al., 2025; Tsai et al., 2020). Multi-level, participatory approaches appear particularly responsive to adolescents' emerging autonomy (Tsai et al., 2020). However, many programs continue to rely primarily on traditional school-based education, yielding short-term knowledge gains but inconclusive behavioral and long-term clinical effects (Peerbhay et al., 2025; Nazari et al., 2025). As interventions are often delivered outside routine clinical care and as standalone initiatives rather than embedded within the care ecosystem, this underscores the importance of structuring clinical communication to extend beyond the dental encounter and resonate in adolescents' everyday oral health practices.

This study is situated in the Norwegian Public Dental Service (PDS), a publicly funded system providing routine dental follow-up for children and adolescents. Despite universal access, two-thirds of 18-year-olds report experience of dental caries, and many develop their first cavity during adolescence (Statistics Norway, 2023). Since the early 1990s, a method for activating patients' motivation for change called motivational interviewing (MI) has been widely adopted to promote healthy behaviors across areas such as physical activity, substance use, gambling, and oral health (Borrelli et al., 2015). Although MI has

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shown promise for supporting behavior change among adolescents, a recent study conducted among dental practitioners in Norway reveals a paradox: a large majority report understanding the theoretical principles of MI, yet only about half feel confident applying it in routine care with adolescent patients (Lassemo et al., 2023).

In response to these challenges, the research and innovation project #Care4YoungTeeth<3 designed the 2x2 concept for adolescent oral health promotion (Høiseth et al., 2024). The concept promotes the simple message of brushing twice daily for two minutes. The four-year project was a multi-stakeholder collaboration involving adolescents, dental professionals, designers, and researchers working through iterative co-design. Its vision was to frame oral care as a companion that follows young people across everyday contexts, rather than a series of isolated instructions delivered in the dental chair.

As part of the 2x2 concept, #Care4YoungTeeth<3 developed the "2x2 hour", an interactive school-based lesson for 8th-grade students designed to extend oral health communication beyond the dental clinic. The "2x2 hour" included several elements, one of which was a goal-setting card. Initial findings from the school-based intervention testing, as well as the rationale for further testing the goal-setting card in a dental clinic setting, are reported elsewhere (Høiseth et al., 2024; Jasbi, 2024).

This paper reports a focused sub-study in which the goal-setting card was redesigned into a coherent three-touchpoint intervention in a dental clinic. Guided by a salutogenic perspective and hope theory, we co-designed and piloted the intervention in collaboration with clinicians. Existing service touchpoints, including appointment messaging and in-clinic dialogue, were reshaped through deliberate attention to relational tone, dialogue structure, and co-authored goal setting. Drawing on interviews with clinicians and adolescent patients, we explore how hope-oriented principles can be operationalized within everyday public dental care.

The paper contributes to healthcare design by showing how service design can support more meaningful clinical communication in public dental care and how hope-oriented principles can be enacted across routine service encounters.

2. A salutogenic and hope-based approach to service design

We adopted a salutogenic lens (Antonovsky, 1979; 1987; 1996) because it shifts attention from reducing risk to strengthening health resources. This perspective reorients health from being understood primarily as the absence of disease to focusing on the factors that support health and well-being (Antonovsky, 1979).

A salutogenic approach is particularly relevant in service design for health as it encourages designers to identify and strengthen everyday resources within existing service encounters (Jasbi & Høiseth, 2023; Jasbi et al., 2024). Salutogenic principles have previously informed architecture and urban planning aimed at creating health-promotive environments (Dilani, 2008). Central to this approach is Antonovsky's concept of sense of coherence, comprising three components: comprehensibility, manageability, and meaningfulness. In adolescent dental care, where consultations are brief and habits are enacted in everyday life, this orientation guided us to design touchpoints that make information clearer (comprehensible), suggested changes feasible (manageable), and oral health personally relevant (meaningful).

Under the salutogenic umbrella, several related concepts are used in health promotion, including quality of life, well-being, sense of coherence, and hopefulness (Haugan & Eriksson, 2021). Hope theory (Snyder, 2000; 2002) complements a salutogenic service design approach by specifying how motivation for change is built through goals, pathways, and agency. In our work, we draw on this to intentionally design and implement hope-oriented elements across the service touchpoints in adolescent oral healthcare.

3. Study context and methods

3.1. Setting

Design researchers in the #Care4YoungTeeth<3 project collaborated with a clinic in the Trøndelag County PDS that had expressed interest in piloting the intervention. The study involved three clinicians from this mid-Norway clinic: one dentist, one dental hygienist, and one dental secretary, each with several years of experience in the Norwegian PDS. The collaboration ran from fall 2024 to May 2025,

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with the clinicians serving as practice-based partners in co-designing, recruiting eligible adolescents and implementing the pilot.

3.2. Co-design process of the three-touchpoint intervention

Building on the previously developed goal-setting card, the three-touchpoint intervention was co-designed with the three collaborating clinicians to ensure contextual suitability.

Co-design sessions were conducted through a series of digital meetings using the online whiteboard tool Miro to explore, visualize, and refine potential touchpoints and their sequencing. Together, we mapped the existing patient journey, identified opportunities for hope-oriented communication, and developed alternative formulations of messages and scripts. Between sessions, the design researchers synthesized input and refined the materials.

Following agreement on a prototype sequence, the goal cards were professionally printed and delivered to the clinic. Information materials and consent forms for eligible adolescents aged 16 years or older, were prepared in parallel, and a brief implementation guide was provided to the participating clinicians.

3.3. A hope-oriented three-touchpoint intervention

Figure 1 illustrates the clinic intervention and its three touchpoints: an SMS appointment invitation sent prior to the consultation, an in-clinic conversation structured around a brushing demonstration and a printed goal card, and a follow-up SMS sent a few weeks after the consultation. Across these touchpoints, salutogenic and hope-oriented principles were applied through tone of voice, dialogue structure, and goal formulation. The standard appointment SMS was adapted to include the relational phrases "Welcome to our place :)" and "Sincerely."

The in-clinic activity formed the core of the intervention. Here, dental clinicians invited adolescents to demonstrate their brushing technique and together identify small, achievable goals. This touchpoint was designed to support comprehensibility by providing clear feedback on current routines, manageability through concrete and feasible steps, and agency through co-authored goal setting. The goal card functioned as a simple communication tool to structure dialogue about oral health routines and document adolescents' own formulations of intended changes. Sized to fit a wallet or phone cover, the card was intended to accompany adolescents into everyday life, and they were encouraged to photograph it to keep it easily accessible. It included a QR code linking to a simple website where adolescents could access additional information, review frequently asked questions, and submit questions anonymously, extending engagement beyond the consultation. The agreed goals were entered into the patient record, and before leaving the clinic, adolescents received a toothbrush and toothpaste. The follow-up SMS was designed to convey relational support, including wording such as "We are still rooting for you and hope the goal on your card helps you take care of your teeth :)" and "A little greeting from all of us."

The intervention deliberately relied on low-fidelity elements to enable delivery with minimal additional effort and cost. No new digital systems or workflows were introduced beyond the existing SMS channel, and the printed card could be used alongside standard clinic materials. The design emphasized tone, timing, and low-threshold elements clinicians could adapt to their style, translating salutogenic and hope-oriented principles into everyday clinical practice.

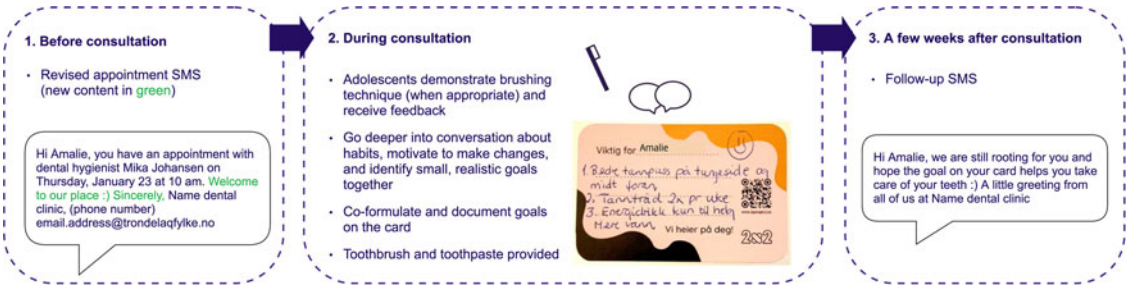


Figure 1. Three-touchpoint intervention. Fictional names. Source: #Care4YoungTeeth<3

3.4. Pilot, research component, and ethics

The pilot explored whether the intervention could help clinicians better understand adolescents' oral health habits, support reflection on everyday routines, and strengthen the link between clinical encounters and daily life. The intervention was delivered to adolescents aged 12–20 years attending

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scheduled recall consultations, typically arranged for patients requiring preventive follow-up due to caries experience or elevated caries risk.
The in-clinic component was piloted over one week in winter 2025 during routine recall consultations, with two clinicians typically present. The follow-up SMS was sent a few weeks after the consultation. For the research component, the aim was to interview three to five adolescents aged 16 years or older, as this group could provide independent consent. Adolescents under 16 could receive the intervention but were not included in the interviews.
During the pilot week, nine eligible adolescents (16+) provided written consent to participate in the research component. This number reflected the limited duration of the pilot and the clinic's capacity within routine care. Due to scheduling constraints and non-response to follow-up contact, three adolescent interviews were completed (two 16-year-old boys and one 18-year-old girl). In addition, one group interview was conducted with the three clinicians involved in the pilot. All interviews were conducted by two researchers (MH and SB) using semi-structured guides. Interviews were conducted digitally via Microsoft Teams, and adolescents joined by phone at their own request (audio only).
As an exploratory qualitative pilot embedded in clinical practice, the study aimed to generate in-depth insights into how the intervention was experienced and enacted, rather than statistically generalizable outcomes. The study was approved by Sikt, the Norwegian Agency for Shared Services in Education and Research.

3.5. Analysis of data

The interview data were analyzed deductively, guided by the hope-oriented theoretical framework outlined above. Substantive content of audio recordings were transcribed into written form. Two researchers independently conducted an initial reading to identify segments related to how the intervention was experienced and enacted. In a second step, statements from clinicians and adolescents were examined in relation to key constructs from hope theory. Interpretations were discussed and refined collaboratively.
In the presentation of findings, adolescents are referred to by pseudonyms to distinguish their perspectives, while clinicians are described at group level, as they were interviewed together and expressed largely overlapping views.

4. Findings

4.1. The appointment SMS as a relational first touchpoint

Dental health personnel described the revised appointment SMS as slightly nicer and more personal. The change was easy to implement through a designated super user of the administrative system. The clinicians also expressed a wish to retain the new formulation: *"We think the text we have had has been okay, but perhaps we should change the standard text – it's a bit nicer now."*
The three interviewed adolescents did not report noticing any difference and described the SMS as *"regular"*.
From a hope-oriented perspective, the revision represents a modest adjustment in tone, shifting the message from a standard appointment notice to a more explicit invitation.

4.2. In-clinic communication: brushing demonstration as support for mastery

The three interviewed adolescents described concrete learning outcomes when invited to demonstrate how they usually brushed their teeth, such as discovering areas they had not cleaned thoroughly and being shown techniques for hard-to-reach areas. Ingrid reflected:

> "*At first, I didn't understand why, but then I realized that it helps me – how I should do it from now on. They told me things I could improve. It was pretty helpful.*"

Receiving feedback directly connected to their own brushing appeared to position the adolescents as active participants in the consultation. The demonstration translated general advice into concrete pathways and supported agency by enabling an experience of *"I can do this,"* thereby supporting a sense of mastery.
The clinicians noted that the demonstration could be time-consuming and occasionally uncomfortable for certain patients. As a result, they used the activity selectively, particularly with younger adolescents or

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when a specific need was evident. This reflects a pragmatic adaptation to time constraints while retaining the demonstration where it was perceived to most effectively support adolescents' sense of mastery.

4.3. In-clinic communication: the goal card as a dialogue tool

When identifying and formulating goals together with adolescents, dental health personnel described a clear conversational sequence: exploring habits and challenges, discussing possible adjustments, and then writing goals on the card. Relationship-building was described as essential and enacted through informal conversation, attentive listening, and mirroring. The clinicians emphasized small, realistic, and individually anchored steps, and noted that adolescents often suggested changes themselves. Goals were formulated through dialogue, with adolescents deciding how they were phrased on the card.

The three interviewed adolescents described different experiences with the goal-setting process. For Ingrid, writing the goal down made it more concrete and visible: *"It was nice to write it down. Then I had it in one place. It became a reminder."* For Sander, the conversation itself appeared more important than the card, which ended up in his pocket without him thinking much about it. He reported having told his parents about the card and showing it to them, but he did not clearly recall how they responded. Still, Sander's account of how the goal was formulated illustrates its role in concretization: *"We formulated that I should get a bit better at brushing properly and on that one side, a bit more often, mouthwash and floss and so on."* He emphasized ownership of the process: *"So that was how I came up with it [the goal]."* Jakob had formulated a goal to switch to an alcohol-free mouthwash and reported implementing the change immediately after the consultation by purchasing a new product the same day. He did not rely on the card itself but noted that it might have been useful had there been more elements to remember. Across these accounts, the card functioned both as a structure for dialogue and as a means of documenting adolescents' own formulations of intended change. One clinician described the relational dimension: *"At the same time, we try to build a relationship, which we know opens up for good communication."* From a hope-oriented perspective, the goal card can be understood as supporting the articulation of personally meaningful goals, specifying concrete pathways, and reinforcing agency through co-authored goal formulation.

4.4. From follow-up to "motivation SMS": extending care beyond the clinic

Dental health personnel began referring to the follow-up SMS as a "motivation SMS" and described sending it as uplifting and professionally meaningful. They experienced it as a personal way to extend care beyond the consultation. One clinician reflected:

> "*The motivation SMS we sent after the consultation was very nice. [It felt] good, as a professional, to be able to send that message. [We] haven't received feedback from the patients, but [I] believe those patients who were here thought it was nice. For instance, [I] wrote 'happy winter break' to those who had one coming*."

The three interviewed adolescents described different reactions to the message. Ingrid experienced it as a helpful reminder: *"Things get a bit washed out. Such [a] message reminded me. [I] have a lot going on in everyday life. [It was] nice to be reminded after a while."* She also noted that the interview itself functioned as a reminder, as no additional messages had been sent. Sander received the SMS while at school and described it as *"a bit fun,"* appreciating it as a sign that the dentist cared. During the interview, he opened the message and read it aloud, commenting, *"[It's] always nice to think that the dentist cares and reminds you, if it's been forgotten."* Jakob described a neutral reaction: *"I didn't feel particularly happy or sad or anything. I just [thought], 'okay'. [I] didn't think more about it."*

From a hope-oriented perspective, the follow-up SMS can be understood as an attempt to reactivate previously formulated goals while signaling continued care.

4.5. Overall impressions of the intervention

In reflecting on the consultation, Sander described the overall experience as less anxiety-provoking than previous visits. He reported *"a bit more motivation to take care of my teeth,"* linking this to the clinician *"being good at showing how to do it."* He also emphasized that seeing his own teeth and the areas that needed improvement made the advice easier to understand. Sander summed up what felt right: *"Meet [youth] with a smile. Don't judge. The way my appointment was [this time] – that's how it should be. Don't change anything."*

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Ingrid described how she had *"made it into a routine"* and was now carrying out the recommended practice every evening. She additionally noted that adolescents may benefit from an extra reminder to follow such advice.
Jakob summarized the visit as *"relatively fine"* and did not clearly recall how it differed from earlier appointments.
From the dental health personnel's perspective, the intervention enabled more personal, two-way communication. They described a structured progression in the conversations, formulating small, realistic, individually tailored goals, and instances where adolescents suggested changes themselves. The goal card functioned as a tool to structure dialogue and document agreed goals. Clinicians expressed a desire to continue using it and considered it particularly useful for adolescents aged 16–18, while the brushing demonstration was prioritized more often for younger teenagers. A recurring theme was a *"good enough"* approach with realistic expectations. As one clinician expressed: *"Floss three times a week. That's better than nothing."*
Two of the three adolescents specifically mentioned receiving a toothbrush and toothpaste and reported using them. The adolescents also referred to touchpoints beyond the clinic, including social media and brief sessions at school, as potential ways to keep oral health messages active between appointments.

5. Discussion

Our study presents a pragmatic setup for co-designing and piloting small service elements within a regulated, resource-constrained clinical context. The findings illustrate how hope-oriented communication can be enacted across appointment messaging, in-clinic dialogue, and follow-up, and how these elements were experienced by clinicians and adolescents. Here, design contributed by reshaping routine clinical interactions into connected touchpoints. In the following, we discuss how support for change manifested across these touchpoints and the practical conditions that made the intervention workable.

5.1. Designing for incremental change within clinical practice

Service design initiatives in healthcare often struggle with long-term implementation and integration into routine workflows (Vink et al., 2021). Rather than introducing ambitious structural change, the three-touchpoint intervention was deliberately designed to operate within existing clinical structures.
Building on earlier phases of #Care4YoungTeeth<3, the co-design process with three clinicians focused on elements that were realistic in scope and compatible with routine care. Iterative digital sessions enabled collaborative refinement of wording, sequencing, and materials. The short pilot period functioned as an early feasibility test, creating space for incremental experimentation within everyday practice. This alignment between design work and operational realities created conditions for continued service improvement, as we observed how the clinicians gradually assumed greater ownership of both the pilot and its touchpoints.
The structured feedback loop, in which we met with the clinician team to share general impressions from the adolescent interviews, further strengthened this process. Clinicians described hearing adolescents' perspectives as motivating. This exchange appeared to validate their effort and sustain engagement with the intervention.
In this way, the contribution of service design lay not primarily in introducing new tools, but in orchestrating coherence across existing touchpoints and translating hope-oriented principles into actionable communication practices within routine care.

5.2. Building hope-oriented coherence across service touchpoints

The intervention comprised three interconnected touchpoints. While clinicians described a clear relational thread across these elements, adolescents' experiences of coherence varied. The appointment SMS did not stand out, whereas the in-clinic interaction and follow-up message were central in adolescents' accounts.
Anchored in a salutogenic perspective (comprehensibility, manageability, meaningfulness) (Antonovsky, 1979; 1987; 1996) and hope theory (goals, pathways, agency) (Snyder, 2000; 2002), the intervention was structured to engage adolescents through related but distinct communicative moments. Each touchpoint carried a specific intention: the SMS to introduce a relational tone; the brushing demonstration to translate advice into concrete action; the goal card to support personally meaningful goal articulation; and the follow-up SMS to reactivate attention after the consultation.

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The three adolescents engaged with these elements differently. For Ingrid, writing the goal enhanced memorability. For Sander, the surrounding dialogue appeared more significant than the card itself. Jakob described the goal-setting process as something that *"just happened,"* suggesting that the intervention at times functioned as a subtle guiding structure rather than a consciously experienced strategy. These variations indicate that hope-oriented communication may operate both explicitly and implicitly, depending on adolescents' readiness for interaction and reflection. They also raise questions about which factors shape how the intervention works, in which situations, and for whom.

In our material, differences appeared partly related to adolescents' openness to reflection. When Jakob was asked what it felt like to formulate a goal, he did not fully recognize the reflective framing of the question. This may relate to how the questions were posed in the interview setting. At the same time, broader factors such as socioeconomic background, health literacy, prior experiences with dental care, and existing oral health routines are likely to influence how such an intervention is received. Even though the intervention was delivered to adolescents attending recall consultations due to caries experience or elevated risk, the small scale of this pilot does not allow conclusions about how it may resonate across different socioeconomic or literacy backgrounds.

Notably, coherence need not be fully recognized by adolescents in order to carry value. Although the appointment SMS did not stand out to the three interviewed adolescents, clinicians experienced it as a meaningful way to signal care and relational intention. Hope-oriented communication may therefore contribute not only to adolescents' experience but also to clinicians' sense of relational engagement within routine care. This asymmetry between design intention and conscious recognition underscores the importance of considering both patient and provider perspectives when assessing relational interventions.

5.3. Selective adaptation within routine care

The intervention combined appointment and follow-up SMS, an in-clinic demonstration, a co-authored goal card, and targeted take-home items (toothbrush and toothpaste). Rather than being applied uniformly, these elements were used selectively in response to clinical judgment. Clinicians prioritized brushing demonstrations when technique-related needs were evident, particularly among younger adolescents, and used the goal card mainly with older adolescents who appeared able to integrate one or two concrete goals into their daily routines. Clinicians described applying a *"good enough"* stance to balance support with time constraints and to avoid unnecessary emotional discomfort. Such selectivity appears to be an important condition for embedding the intervention within routine care, allowing hope-oriented elements to remain viable within everyday workflow rather than becoming an added burden.

Extending care beyond the consultation required minimal additional infrastructure. Existing SMS systems were used with minor wording adjustments, and goal cards were easily stored and distributed. The take-home items supported translation of advice into action for two of the adolescents interviewed, who explicitly reported using them. No new digital platforms or longer consultations were required; additional time commitments were specific to research procedures rather than the intervention itself.

In this sense, the intervention appeared workable because clinicians could adjust and prioritize elements rather than deliver them as a fixed package. Adolescents also pointed to channels beyond the clinic, particularly social media and brief school-based sessions, as opportunities to sustain attention between consultations. In combination with the "2x2 hour" developed in #Care4YoungTeeth<3, the intervention can thus be understood as one modest step toward framing oral care as a companion that accompanies adolescents across everyday contexts rather than remaining confined to the dental chair.

5.4. Transfer to routine clinical practice

Our findings suggest a transferable pattern for routine clinical practice: small, selective design elements embedded within existing workflows rather than added as separate initiatives. The intervention combined multiple touchpoints, recognizing that different elements resonate differently among adolescents. Adaptation therefore requires attention to both patient needs and clinicians' operational practices.

What may be transferable to other healthcare settings is less the specific tools and more the service design logic: aligning touchpoints through a shared relational intention, keeping elements low-threshold, and accepting selective use based on clinical judgment. Brief journey mapping, aligned wording, and simple guidelines for prioritizing can help preserve coherence without increasing burden on healthcare staff.

At a closing seminar, the clinician team presented the pilot and reflected on its trade-offs. The way in which they shared their experiences suggested that internal dissemination can strengthen ownership and

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interdisciplinary learning. In turn, transfer may be more feasible when it builds on existing IT channels and communication practices. Concise templates, light coordination, and short feedback loops can support gradual adaptation across sites.

The choice of SMS as communication channel may appear counterintuitive, as SMS is not necessarily adolescents' preferred medium in their everyday social communication. However, within Norwegian public dental care, SMS constitutes the established and regulated channel for appointment messaging. A central design premise of this intervention was to avoid introducing new platforms, extended consultations, or additional system complexity. Working within existing infrastructure allowed relational adjustments to be implemented with minimal added burden and cost.

Beginning with limited pilots allows healthcare services to test feasibility and organizational fit before pursuing broader implementation. Assessing clinical effectiveness in terms of oral health outcomes would require larger studies with longer follow-up and relevant behavioral or clinical indicators.

5.5. Limitations

This study draws on a small pilot conducted in one particularly motivated clinic and included only a few adolescent participants. The findings should therefore be understood as context-bound and exploratory. The intervention may function differently in other clinics, with other practitioners, or among adolescents with different socioeconomic backgrounds, levels of engagement, or health literacy.

6. Conclusion

Our study shows how a hope-based, salutogenic perspective can be translated into a simple three-touchpoint intervention within public dental care. This exploratory pilot suggests that carefully designed service elements can support reflective clinical conversations and help adolescents connect clinic experiences with everyday oral health routines. At the same time, further research in broader and more varied settings is needed to examine how such interventions contribute to oral health promotion across contexts and populations.

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